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Automated Detection of Patient Ventilator Asynchrony Using Pes Signal A Feasibility Study Towards a Detection Algorithm

NCT06186557 NCT

Recruiting Observational

Location
Netherlands

Trial date
01 Feb 2023 - 30 Apr 2025

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External ID

NCT06186557

Other IDs

2202-061

Conditions

Mechanical Ventilation Complication

Keywords

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Trial description

INTRODUCTION AND RATIONALE Mechanical ventilation should unload the respiratory muscles, provide adequate gas exchange and should be safe, i.e., harm due to mechanical ventilation should be reduced to a minimum. To achieve this the interaction between the ventilator and the patient is preferentially synchronous. Ventilator settings not being synchronized with patient respiratory drive or activity is a phenomenon known as patient-ventilator asynchrony (PVA). PVA may induce several deleterious effects.1 Studies have shown asynchronies to be associated with patient discomfort, increased work of breathing, prolonged weaning, and in one study, even increased mortality.

Monitoring PVA however is difficult. Clinicians often have to rely on physical examination of the patient as well as visual inspection of pressure, flow and volume waveforms to identify an asynchrony.6 The sensitivity and positive predictive value of analyzing breath-to-breath waveforms are very low (22% and 32%, respectively).1 Artifacts such as cardiac oscillation may mimic asynchronies, and there are times when clinicians standing at the bedside are unable to distinguish between asynchronies and artifacts with certainty.6 Furthermore, detection of PVA is dependent of bedside examination. This challenge leads to the desire of developing effective automated PVA recognition algorithms.7 Various automated algorithms have been developed, however with a variable performance.1 For a correct analysis of asynchronies, the use of an esophageal balloon catheter, which measures the esophageal pressure (Pes), or a catheter which measures the electrical activity of the diaphragm, is necessary.1 Since the use of Pes catheters, it is possible to describe other forms of PVA, such as reverse triggering, which is an asynchrony in which the ventilator triggers the patient.8 Until recently, however, the esophageal catheter has not been routinely used in daily practice but more as a research tool. Since the introduction of personalized medicine, clinicians have gained interest in esophageal manometry to better titrate care to the unique physiology of a patient.9 There are in the current literature no reports of PAV detection algorithms that use the Pes signal for detection.

In the LUMC the investigators use the esophageal catheter in all patients admitted with acute respiratory failure and in patients ventilated for more than 48 hours per protocol. The esophageal signal gives the opportunity to detect asynchronies more easily than without. The investigators

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OBJECTIVES 2.1 Primary Objective The primary objective of this study is to develop an asynchrony detection algorithm based on pressure, flow and Pes signals of patient data using ML.

2.2 Secondary Objectives Secondary objectives are to validate the detection algorithm by comparing its performance with the assessment of the expert panel and to compare its performance with the performance of a second algorithm which is based on pressure- and flow signals only.

STUDY DESIGN This study will take place at the Intensive Care Unit of the LUMC. First, internal data of adult ICU patients on mechanical ventilation because of acute respiratory failure or with a ventilation duration of at least 24 hours and that are equipped with an esophageal balloon catheter will be collected from the ventilator. The data of interest include pressure, flow and Pes signals of the ventilation. It is necessary to collect as much data as possible as this is required for the development of the algorithm. A minimum of 50 patients will be included with from each a ventilation recording between 4 and 8 hours, which amounts to 200 - 400 hours of mechanical ventilation recording.

The following labels will be assigned to the data:

Trigger asynchrony (early (reversed), false, failed)

Cycle (early, late)

Double trigger (combination of trigger and cycle problem)

The data will be checked for regions of interest. Regions of interest are regions in which a lot of

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a asynchrony label than the breath cycle will be included for construction of the model. An equivalent amount of normal breaths will also be used to train the algorithm.

The data will be separated in two datasets. The first set will contain the first half of the patients, i.e. the patients with an odd research number. The second set will contain the second half of the patients (patients with an even research number). Research numbers are allocated consecutively to patients after consent.

The machine learning (ML) technique that will be used is a Convolutional Neural Network which will be developed in cooperation with the Technical University of Delft (Technical Medicine and Dr. D.M.J. Tax). The algorithm will be developed based on this training data and the purpose is that the algorithm learns to generalize from the training set.

The algorithm will be validated on the test data for the detection of PAV. The test data will also be annotated by the expert panel. The algorithm's detection performance will be compared with the annotation of the expert panel on the same data.

Together with the development of the main algorithm, which is based on three signals (flow, pressure and esophageal pressure (Pes)), another algorithm based on two signals (flow and pressure) will be developed with the same AI technique. This second algorithm has most in common with previous described algorithms in the current literature. The performance of the first algorithm will be compared with the second algorithm using the test data (second part of the data) in order to investigate if addition of the Pes signal is superior in the detection of asynchronies.

Eligibility

Demographics

Minimum Age

18

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100

Inclusion Criteria

- admission to the ICU of the LUMC;
- age of 18 years or older;
- intubated and receiving mechanical ventilation because of acute respiratory failure or with a ventilation duration of at least 24 hours; and
- equipped with an esophageal balloon catheter

Exclusion Criteria

- after recent pneumectomy or lobectomy;
- no informed consent

Interventions 1

No intervention

Other

No intervention

Primary outcomes 1

Performance of detection algorithm

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algorithm will be compared with the labels of the experts to assess the performance of the algorithm. The performance of the primary algorithm will be compared with the performance of the second algorithm, which is based only on pressure and flow signals. The performance of the second algorithm will be assessed as described above. The agreement between the experts will be assessed using Fleiss's kappa, which evaluates the agreement between more than two raters.

Trial information

Enrollment count

50

Last update

13 Mar 2024

Organization

Leiden University Medical Center

Sponsor

Leiden University Medical Center

Responsible

Abraham Schoe, MD, PhD. , PRINCIPAL_INVESTIGATOR
MD, PhDLeiden University Medical Center

Investigators

Abraham Schoe, MD, PhD
Principal Investigator, Leiden University Medical Centre

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1

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Read the information to understand more what the trial is about. In some cases, there might be a brochure with more information on this clinical trial.

2

Step 2: Check for a local clinical trial

Visit the Participating Locations to check if the trial is conducted in your country.

3

Step 3: Establish direct Contact

If there are contact details, directly reach out to the trial coordinator for more guidance.

No contact details? Click on this contact form for assistance in obtaining the necessary information or discuss with your physician for the best course of action.

4

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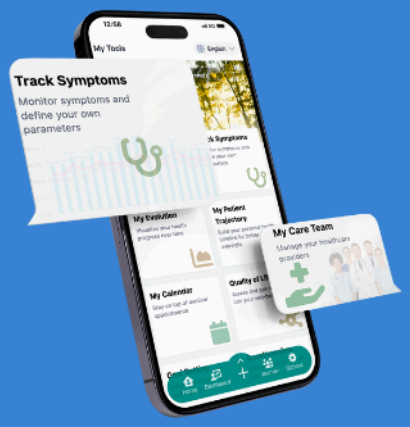

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Eligibility Criteria

Sex
All ✓

Age
18 Years to 100 Years

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Participating locations

Netherlands

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